Propositional Resolution

$$\frac{A \vee L \qquad \neg L \vee B}{A \vee B}$$

Soundness proof:

- Let e be an interpretation in which both $e(A \vee L) = 1$ and $e(\neg L \vee B) = 1$
- if $e(L) = 1$ then from $e(\neg L \vee B) = 1$ we conclude $e(B) = 1$, so $e(A \vee B) = 1$
- if $e(L) = 0$ then from $e(A \vee L) = 1$ we conclude $e(A) = 1$, so $e(A \vee B) = 1$
- In any case $e(A \vee B) = 1$.

Propositional Resolution on Clauses

Rule on formulas:

$$\frac{A \vee L \qquad \neg L \vee B}{A \vee B}$$

When we represent disjunctions as sets of literals becomes:

$$\frac{A \cup \{L\} \qquad \{\neg L\} \cup B}{A \cup B}$$

To prove that a formula is valid, we prove that its negation is contradictory by deriving an empty clause (which represents false).

Example Proof of Contradiction by Resolution

$\{\neg p, \neg q, r\}$ $\{\neg q, p\}$ $\{q, r\}$ $\{\neg r\}$

Example Proof of Contradiction by Resolution

$$\begin{tikzcd}
\{\neg p, \neg q, r\} \arrow[dr] & & \{\neg q, p\} \arrow[dl] & \{q, r\} & \{\neg r\} \\
& \{\neg q, r\} & & &
\end{tikzcd}$$

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& & \{r\} & & & &
\end{tikzcd}
$$

Example Proof of Contradiction by Resolution

$$
\begin{tikzcd}[column sep=small]
\{\neg p, \neg q, r\} \arrow[dr] & & \{\neg q, p\} \arrow[dl] & & \{q, r\} \arrow[dd] & \{\neg r\} \arrow[ddl] \arrow[ddddl] \\
 & \{\neg q, r\} \arrow[ddrr] & & & & \\
 & & & & \{q\} \arrow[dl] & \\
 & & & \{r\} \arrow[dr] & & \\
 & & & & \varnothing &
\end{tikzcd}
$$

Consistency by Absence of Contradiction

Conversely, if the set is contradictory, then existentially quantifying over all variables yields false, so applying resolution exhaustively also yields false.
Resolution is **complete**.
Therefore, if resolution does not yield false, the set is consistent.

$$
\begin{tikzcd}
\{\neg p, \neg q, r\} \arrow[d] & \{\neg q, p\} \arrow[dl] & \{q, r\} \arrow[ddll] & \\
\{\neg q, r\} \arrow[d] & & & \text{(resolve all on p)} \\
\{r\} & & & \text{(resolve all on q)}
\end{tikzcd}
$$

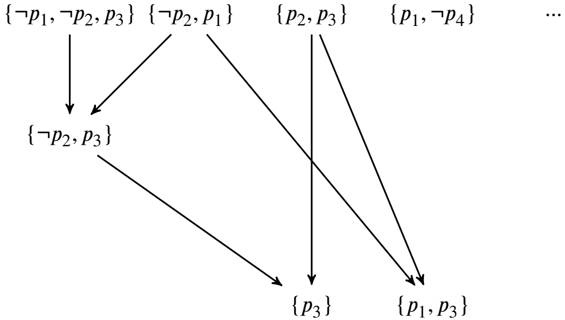
Further resolution attempts would only yield clauses that are subsumed (their subsets, which are stronger, are already derived). No empty clause is generated, so the original set is consistent (a model: $p \mapsto 1, r \mapsto 1$)

Compactness

Infinite set of Formulas

Suppose that we have a countably infinite set of formulas, with countably many propositional variables

Apply resolution exhaustively to larger and larger prefixes of this infinite set



Suppose we are not finding a contradiction in such way. Is the entire infinite set consistent?
Equivalently: if a countable set is contradictory, is there always a finite subset that is contradictory (i.e. if the set is contradictory, can contradiction be found in finite time)?

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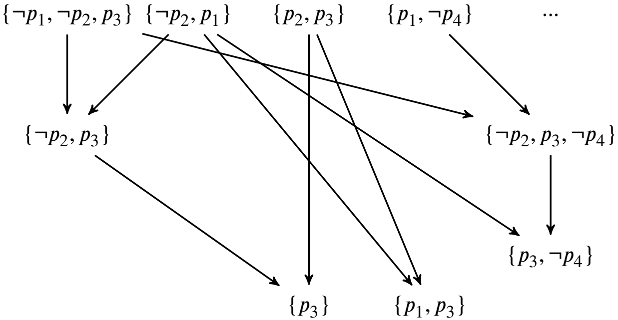

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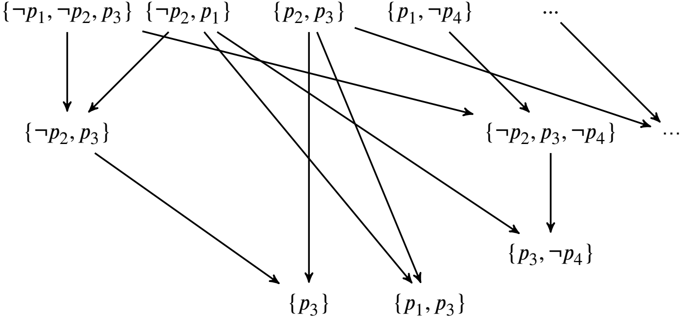

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Compactness

Theorem (Compactness for Propositional Logic.)

Let S be a set of propositional formulas.
Then S is satisfiable iff every finite subset of S is satisfiable.

Equivalently: S is contradictory iff some finite subset of S is contradictory.

Remark: Compactness is a non-trivial property. In propositional logic with infinite disjunctions (which we will generally not consider) it would not hold. In such *infinitary logic* we could take $S = \{D, p_1, p_2, p_3, p_4, \ldots\}$ where

$$D = \bigvee_{i=1}^{\infty} \neg p_i$$

that is, D is equivalent to $\exists i. \neg p_i$. In this example, every finite subset of S is satisfiable. For example, $\{D, p_1, p_2, p_3\}$ is satisfied by interpretation that sets p_1, p_2, p_3 to 1 and p_4 to 0. But S itself is not satisfiable: D requires $e(p_i) = 0$ for at least one i, whereas $p_i \in S$.

Proof of Compactness

One direction is trivial: if S is satisfiable then there exists e such that $e(S) = 1$. By definition, this means $e(F) = 1$ for every $F \in S$. Then for every finite subset $T \subseteq S$ we have $e(T) = 1$, so T is satisfiable. So, the point is to show the converse.
Intuition: A finitely satisfiable set has "all finite pieces" satisfiable (using potentially different interpretations e). The question is whether we can somehow assemble interpretations for all finite pieces T into one large interpretation for the entire infinite set S. We will define such interpretation by extending it, variable by variable, while preserving finite satisfiability for the interepretations that begin with values for propositional variables chosen so far.
Let every finite subset S be satisfiable. Let $p_1, p_2, \ldots$ be the sequence of all propositional variables for formulas in S. Let $V = \{p_1, p_2 \ldots, \}$ (this set is countable by our assumption on syntax of formulas, but can be infinite).
Given a sequence of boolean values $u_1, u_2, \ldots, u_n \in \{0, 1\}$ of length $n \geq 0$, by an $(u_1, u_2, \ldots, u_n)$-interpretation we mean an interpretation $e : V \to \{0, 1\}$ such that $e(p_1) = u_1, \ldots, e(p_n) = u_n$.

Proof: Constructing Interpretation

We will define interpretation $e^*(p_k) = v_k$ where the sequence of values $v_1, v_2, \ldots$ is given as follows:

$$v_{k+1} = \begin{cases} 0, & \text{if for every finite } T \subseteq S, \text{ there exists a} \\ & (v_1, \ldots, v_k, 0) - \text{interpretation } e \text{ such that } e(T) = 1 \\ 1, & \text{otherwise} \end{cases}$$

We next show by induction the following.

FIRST PART.
Claim: For every non-negative integer k, every finite subset $T \subseteq S$ has a $(v_1, \ldots, v_k)$-interpretation e such that $e(T) = 1$.
Base case: For $k = 0$ the statement reduces to the claim that every finite subset of S is satisfiable, which is an assumption of the theorem.

Inductiveness and the Model

Inductive step: Assume the claim for k: every finite subset $T \subseteq S$ has a $(v_1, \dots, v_k)$-interpretation e such that $e(T) = 1$, we show that the statement holds for $k + 1$.
If $v_{k+1} = 0$, the inductive statement holds by definition of v_{k+1}. Let $v_{k+1} = 1$.
Then by definition of v_{k+1}, there exists a finite set $A \subseteq S$ that has no $(v_1, \dots, v_k, 0)$ interpretation. We wish to show that every finite set $B \subseteq T$ has a $(v_1, \dots, v_k, 1)$-interpretation e such that $e(B) = 1$. Take any such set B. Consider the set $A \cup B$. This is a finite set, so by inductive hypothesis, it has a $(v_1, \dots, v_k)$-interpretation, say e, where $e(A \cup B) = 1$. Because $e(A) = 1$, which has no $(v_1, \dots, v_k, 0)$-interpretation, we have $e(p_{k+1}) = 1$. Therefore, e is a $(v_1, \dots, v_k, 1)$-interpretation for $A \cup B$, and therefore for B. This completes the inductive proof.

From Sequence of Interpretations to One

We have shown that for every non-negative integer k, every finite subset $T \subseteq S$ has a $(v_1, \ldots, v_k)$-interpretation e such that $e(T) = 1$.
We now define $e^*(p_k) = v_k$ for all k.

Observe that, for every M, e^* is a $(v_1, \ldots, v_M)$-interpretation.

SECOND PART.
We finally show that $e^*(S) = 1$. Let $F \in S$. Let $FV(F) = \{p_{i_1}, \ldots, p_{i_k}\}$ and $M = \max(i_1, \ldots, i_k)$. Then $FV(F) \subseteq \{p_1, \ldots, p_M\}$. The singleton set $\{F\}$ is finite, so, by the Claim, it has a $(v_1, \ldots, v_M)$-interpretation e such that $e(F) = 1$. Because e^* is also a $v_1, \ldots, v_M$-interpretation, and it agrees with e on all variables in F, we have $e^*(F) = e(F) = 1$.

We have therefore shown that $e^*(F) = 1$ for all $F \in S$, so $e(S) = 1$, as desired.

Why did this work?

How does this proof break if we allow infinite disjunctions? Consider the above example $S = \{D, p_1, p_2, p_3, \ldots\}$ where $D = \bigvee_{i=1}^{\infty} \neg p_i$. The inductively proved claim still holds, and the sequence defined must be $1, 1, 1, \ldots$. Here is why the claim holds for every k. Let k be arbitrary and $T \subseteq S$ be finite. Define

$$m = \max(k, \max\{i \mid p_i \in T\})$$

Then consider interpretation that assigns to true all p_j for $j \leq m$ and sets the rest to false. Such interpretation makes D true, so if it is in the set T, then interpretation makes it true. Moreover, all other formulas in T are propositional variables set to true, so the interpretation makes T true. Thus, we see that the inductively proved statement holds even in this case. What the infinite formula D breaks is the second part, which, from the existence of interpretations that agree on an arbitrarily long finite prefix derives an interpretation for infinitely many variables. Indeed, this part explicitly refers to a finite number of variables in the formula.

Term Models for First-Order Logic

Example Formula in First-Order Logic

model of a formula = interpretation (structure) that makes a formula true

$$
\begin{aligned}
\neg\big(\quad & (\forall x.\exists y.\ R(x,y)) \land \\
& (\forall x.\forall y.\ (R(x,y) \rightarrow \forall z.\ R(x, f(y,z)))) \land \\
& (\forall x.\ (P(x) \lor P(f(x,a)))) \\
& \rightarrow \forall x.\exists y.\ (R(x,y) \land P(y))\big)
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After normal form and Skolemization we obtain these first-order clauses:

$$
\begin{aligned}
& R(x, g_1(x)) \\
& \neg R(x,y) \lor R(x, f(y,z)) \\
& P(x) \lor P(f(x,a)) \\
& \neg R(c_0, y) \lor \neg P(y)
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- variables are implicitly $\forall$ quantified; there are no $\exists$ quantifiers
- each clause is disjunction of literals (atomic formulas or their negation)
- from any model of these clauses we can obtain model for the original formula (just ignore interpretation of Skolem constants g_1, c_0)

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$$\neg\Big(\begin{array}{l} (\forall x.\exists y.\ R(x,y)) \land \\ (\forall x.\forall y.\ (R(x,y) \to \forall z.\ R(x,f(y,z)))) \land \\ (\forall x.\ (P(x) \lor P(f(x,a)))) \\ \to \forall x.\exists y.\ (R(x,y) \land P(y)) \end{array} \Big)$$

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Do given universally quantified formulas have a model?

Remark on Notation

We can view n ary relation on D either as a function

$$r_f : D^n \to \{0, 1\}$$

which is how we defined it, so far, or, more conventionally, as the subets

$$r \subseteq D^n$$

through isomorphism such that

$$(d_1, \ldots, d_n) \in r \quad \Longleftrightarrow \quad r_f(d_1, \ldots, d_n) = 1$$

In the following slides we will use the view $r \subseteq D^n$

Finding a Smaller Model

Small model theorems in logic: "if a given set of formulas has a model, then it has a model of a particular kind (e.g. small)"

- First place to look for smaller models: **substructures**

Given a structure (interpretation) (D, α) a substructure is (D', α') where

- $D' \subseteq D$
- for elements in D', α' defines the relations and functions in the same way, so $\alpha'(R) = \alpha(R) \cap (D')^n$ for $n = ar(R)$, and $\alpha'(f)(x_1, ..., x_n) = \alpha(f)(x_1, ..., x_n)$ for $n = ar(f)$
- (D', α') is a valid interpretation, in particular, it maps function symbols of arity n to total functions on $(D')^n \to D'$

Observation: Given (D, α), a substructure is uniquely given by its domain $D' \subseteq D$. The domain D' defines a substructure if and only if it is closed under the interpretation of all function symbols f:

$$\bigwedge_{f \in L_F} \forall x_1, ..., x_n \in D'. \ \alpha(f)(x_1, ..., x_n) \in D'$$

Examples of Substructures

$L = \{f, a, b, T\}$ where

- f, a, b are functions symbols of arity $2, 0, 0$, respectively; $L_F = \{f, a, b\}$
- T is a binary relation symbol

(D, α) is given by $D = \mathbb{R}$ (real numbers) and

- $\alpha(a) = 0$, $\alpha(b) = 1$
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How do substructures look like?

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- Then the set $D'_2 = \{0, 1, 2\}$ does not form a substructure because it is not closed under addition, e.g. $1 + 2 \notin D'_2$.
- The set of integers $D'_3 = \mathbb{Z}$ induces a substructure because: (i) $\alpha(a) \in \mathbb{Z}$, (ii) $\alpha(b) \in \mathbb{Z}$, and (iii) $x, y \in \mathbb{Z} \to x + y \in \mathbb{Z}$.

Universal Formulas Stay True in Substructures

Consider a **universal** formula, with only universal quantifiers (e.g. after Skolemization)

$$\forall x_1, ..., x_n.\ G(x_1, ..., x_n)$$

where G is quantifier free. Suppose this formula is true in (D, α). This means

$$\forall d_1, ..., d_n \in D.\ [\![G(x_1, ..., x_n)]\!]^{\alpha[x_i:=d_i]_{i=1}^n}$$

Let (D', α) be a substructure of (D, α). Then from $D' \subseteq D$ follows also

$$\forall d_1, ..., d_n \in D'.\ [\![G(x_1, ..., x_n)]\!]^{\alpha'[x_i:=d_i]_{i=1}^n}$$

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Theorem

If a set of universal first-order formulas is true in a structure, then it is true in each of its substructures.

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Our goal: find a small substructure

Smallest Substructure

(D, α) is given by $D = \mathbb{R}$ (real numbers) and

- $\alpha(a) = 0,\ \alpha(b) = 1$
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Least fixpoint of function $H(D_k) = \{\alpha(a), \alpha(b)\} \cup \{\alpha(f)(x, y) | x, y \in D_k\}$

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Let D' be a substructure. Which elements must it contain?

- $0,\ 1$ (interpretations of constants)
- $0 + 1,\ 1 + 0,\ 1 + 1$ (adding up constants), so $2 \in D'$
- $2 + 1 = 3 \in D'$
- every non-negative integer

Define: $D_0 = \emptyset,\ D_{i+1} = \{0, 1\} \cup \{x + y | x, y \in D_i\}$ i.e.
$D_{i+1} = \{\alpha(a), \alpha(b)\} \cup \{\alpha(f)(x, y) | x, y \in D_i\}$. Let $D^* = \bigcup_{i \geq 0} D_i$

Least fixpoint of function $H(D_k) = \{\alpha(a), \alpha(b)\} \cup \{\alpha(f)(x, y) | x, y \in D_k\}$ Evey set D_i is finite.
D^* is countable: can enumerate elements of D_1, followed by the elements of D_2, D_3,...
establishing bijection with $\mathbb{N}$

Definition of Smallest Substructure

Language L with function symbols $L_F \subseteq L$.

$$D_0 = \varnothing$$

$$D_{i+1} = \bigcup_{f \in L_F} \{\alpha(f)(x_1, ..., x_n) | x_1, ..., x_n \in D_i\}$$

$$D^* = \bigcup_{i \geq 0} D_i$$

Note: D_i for $i \geq 1$ includes the interpretations of all constants, which are functions of arity $n = 0$

Theorem

- *D^* is the domain of the smallest substructure of (D, α)*
- *D^* is*
    - *always countable*
    - *non-empty $\leftrightarrow$ L contains at least one constant symbol*
    - *finite when L has no function symbols except for constants*

Countable Model Theorem

Lemma

*A set of **universal** first-order formulas has a model if and only if it has a countable model.*

Proof.

Let (D, α) be a model. Then D^* induces a countable sub-structure. Because all formulas are universal, they remain true in D^*. □

Theorem (Downward Löwenheim-Skolem)

A set of first-order formulas has a model if and only if it has a countable model.

Proof.

Let the set of formulas have a model. Transform the formulas into normal form and skolemize them to eliminate existential quantifiers, which introduces a countable number of skolem functions. Then there is a model for the resulting set of universal formulas as well. By previous lemma, then there is also a countable model. Ignoring the interpretation of Skolem constants, we obtain a countable model for the original formula. □

Example: Dense Orders

Consider these axioms, which define *dense linear orders* without upper bound:

$$
\begin{array}{l}
\forall x.\neg T(x,x) \\
\forall x \forall y \forall z. T(x,y) \land T(y,z) \rightarrow T(x,z) \\
\forall x \forall y. (T(x,y) \rightarrow \exists z.(T(x,z) \land T(z,y))) \\
\forall x \exists y. T(x,y)
\end{array}
$$

Real numbers with strict inequality $<$ interpreting relation symbol T are a model of these axioms. Find one countable non-empty model using our construction.

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$$\begin{aligned}
&\forall x. \neg T(x,x) \\
&\forall x \forall y \forall z. T(x,y) \wedge T(y,z) \to T(x,z) \\
&\forall x \forall y. (T(x,y) \to \exists z. (T(x,z) \wedge T(z,y))) \\
&\forall x \exists y. T(x,y)
\end{aligned}$$

Real numbers with strict inequality $<$ interpreting relation symbol T are a model of these axioms. Find one countable non-empty model using our construction.
Skolemizing the existential quantifier for density using $g(x,y)$ and for no-bound with $h(x)$:

$$\begin{aligned}
&\neg T(x,x) \\
&\neg T(x,y) \vee \neg T(y,z) \vee T(x,z) \\
&\neg T(x,y) \vee (T(x,g(x,y)) \wedge T(g(x,y),y)) \\
&T(x,h(x))
\end{aligned}$$

Finding Non-Empty Countable Model

$$\neg T(x,x)$$
$$\neg T(x,y) \vee \neg T(y,z) \vee T(x,z)$$
$$\neg T(x,y) \vee (T(x,g(x,y)) \wedge T(g(x,y),y))$$
$$T(x,h(x))$$

Theorem ensures we can find interpretation of g, h.
One possibility:

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Theorem ensures we can find interpretation of g, h.
One possibility: $g(x, y) = (x + y)/2 \quad h(y) = y + 1$
Since we have no constant and do not wish to have an empty domain, just pick any element as the starting point.

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Apply closure under operations. Here they are all Skolem operations, but in general we use all operations we have, original or Skolem. Describe the set generated in this way.

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Answer: The set of all non-negative numbers representable in binary notation $\overline{b_1 ... b_p . d_1 ... d_q}$, that is:

$$\left\{ \frac{p}{2^k} \mid p, k \in \mathbb{N} \right\}$$

Note that this is a countable set.

Finding Non-Empty Countable Model

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Note that this is a countable set. Try also $g(x, y) = x + 1/(1 + y - x)$

Herbrand (Term) Model: A Generic Countable Model

Instead of looking at arbitrary countable domains and functions on them, we show we can consider a more special class of structures: *ground term models*.
In these models the domain the set of expressions (group terms) built from constants and function symbols, and operations as just constructors.
Remember (D, α) is given by $D = \mathbb{R}$ (real numbers) and

- $\alpha(a) = 0,\ \alpha(b) = 1$
- $\alpha(f)(x, y) = x + y$
- $\alpha(T) = \{(x, y) | x \leq y\}$

The smallest substructure is given by $D_0 = \emptyset,\ D_{i+1} = \{0, 1\} \cup \{x + y | x, y \in D_i\},\ D^* = \bigcup_{i \geq 0} D_i$.
This is precisely the set of values of all expressions built from $0, 1$ and $+$.
In general, the least substructure is the set of values of ground terms:

$$D^* = \{[\![t]\!]^\alpha \mid t \in GT_L\}$$

GT_L is the set of all ground terms (terms without variables) in language L

Values of Ground Terms Induce Smallest Substructure

GT_L is the least set such that if $f \in L$, $ar(f) = n$ $(n \geq 0)$ and $t_1, ..., t_n \in GT_L$ then $f(t_1, ..., t_n) \in GT_L$.
In other words, define $GT^0 = \varnothing$ and

$$GT^{i+1} = \{f(t_1, ..., t_n) \mid f \in L \wedge t_1, ..., t_n \in GT^i\}$$

Then the set of all ground terms is $\bigcup_{i \geq 0} GT^i$

- GT^i is the set of terms of height (depth) at most $i - 1$

Compare to: $D_0 = \varnothing$, $D_{i+1} = \bigcup_{f \in L_F} \{\alpha(f)(x_1, ..., x_n) | x_1, ..., x_n \in D_i\}$

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By induction we prove easily

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Therefore, $D^* = \{[\![t]\!]^\alpha \mid t \in GT_L\}$

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$$D_i = \{[\![t]\!]^\alpha \mid t \in GT^i\}$$

Therefore, $D^* = \{[\![t]\!]^\alpha \mid t \in GT_L\}$
How to define meaning of $f \in L$ as function $GT_L^n \to GT_L$

Interpreting Functions on Ground Terms

Given a language L we are defining an interpretation (GT_L, α_H). If there are no constants, invent a fresh constant a_0 and add it into L.
For function symbols f, we just let

$$\alpha_H(f)(t_1, ..., t_n) = f(t_1, ..., t_n)$$

because we can always build a larger term.
This definition does not depend on the original model (D, α).

We next want to define $\alpha_H(R)$ for each relation symbols $R \in L$
Idea: define the truth value following the truth value in (D, α)

$$\alpha_H(R) = \{(t_1, ..., t_n) \mid ([\![t_1]\!]^\alpha, ..., [\![t_n]\!]^\alpha) \in \alpha(R)\}$$

To determine if relation holds on ground terms, just check if it holds on their values.
It is in this step that we used the original structure (D, α) to define the new structure (GT_L, α_H). We postponed evaluation to relations.

Revisiting Example of Dense Orders

$$\neg T(x,x)$$
$$\neg T(x,y) \vee \neg T(y,z) \vee T(x,z)$$
$$\neg T(x,y) \vee (T(x,g(x,y)) \wedge T(g(x,y),y))$$
$$T(x,h(x))$$

Use the model $(\mathbb{R},\alpha)$ in which T is $<$, $g(x,y)=(x+y)/2$, $h(y)=y+1$ to define Herbrand model (GT_L,α_H). Add fresh constant c.
Define

- $\alpha_H(c)$
- $\alpha_H(g)$
- $\alpha_H(h)$
- $\alpha_H(T)$

Example: why a formula holds in the ground model

Now use this definition of $\alpha_H(T)$.
Take any formula, say

$$\neg T(x,y) \vee (T(x,g(x,y)) \wedge T(g(x,y),y))$$

We wonder if it holds in (GT_L, α_H). Let $x, y, z \in GT_L$. Say $x = c$, $y = h(c)$. Why does

$$\neg T(c,h(c)) \vee (T(c,g(c,h(c))) \wedge T(g(c,h(c)),h(c)))$$

hold?

Example: why a formula holds in the ground model

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Take any formula, say

$$\neg T(x,y) \vee (T(x,g(x,y)) \wedge T(g(x,y),y))$$

We wonder if it holds in (GT_L, α_H). Let $x, y, z \in GT_L$. Say $x = c$, $y = h(c)$. Why does

$$\neg T(c,h(c)) \vee (T(c,g(c,h(c))) \wedge T(g(c,h(c)),h(c)))$$

hold?
Because the same formula holds in the original structure. We defined $[\![T]\!]^{\alpha_H}$ so that

$$(c,h(c)) \in [\![T]\!]^{\alpha_H} \quad \leftrightarrow \quad ([\![c]\!]^{\alpha}, [\![h(c)]\!]^{\alpha}) \in [\![T]\!]^{\alpha}$$

Herbrand Model is a Model of Same Universal Formulas

Lemma

For every quantifier-free formula $G(x_1, ..., x_n)$, if $\alpha_H(x_i) = t_i$ then

$$[\![G(x_1, ..., x_n)]\!]^{\alpha_H} \leftrightarrow [\![G(x_1, ..., x_n)]\!]^{\alpha[x_i := \alpha(t_i)]_{i=1}^n}$$

Proof by induction, using the definition of $\alpha_H(R)$ in the base cases.

Theorem (Herbrand)

Let (D, α) be a model of a set S of universal first-order formulas in the language L containing at least one constant. Then (GT_L, α_H) is also a model of these formulas.

Proof. Let $F \in S$ be of the form $\forall x_1, ..., x_n.G(x_1, ..., x_n)$. Then F holds in (D, α). Let $t_1, ..., t_n \in GT_L$ be arbitrary. Then by the above lemma,

$$[\![G(x_1, ..., x_n)]\!]^{\alpha_H[x_i := t_i]_{i=1}^n} \leftrightarrow [\![G(x_1, ..., x_n)]\!]^{\alpha[x_i := \alpha(t_i)]}$$

Last formula is true because F holds in (D, α). So, F holds in (GT_L, α_H).

Viewing Herbrand Model as Propositional Model

Set S of universal formulas. Suppose we write universal variables as free variables. There is a model (D, α) if and only if there is Herbrand model (GT_L, α_H).
How do we check if a set S has some Herbrand model? Function symbol interpretations are fixed. Need to check if there exists interpretation of each relation symbol R such that

$$\forall G \in S. \forall t_1, ..., t_n \in GT_L. \ [\![G[x_1 := t_1, ..., x_n := t_n]]\!]^{\alpha_H} = true$$

Expand all these universal quantifiers:

$$S' = \{G[x_1 := t_1, ..., x_n := t_n] \mid G \in S\}$$

Then S holds in GT_L if and only if S' holds in GT_L. We have countable domain GT_L and allow countable sets, so we instantiated.
S' has no variables, so it is like a propositional model.

Propositions with Long Names

For each relation symbol R define Herbrand atoms (ground instances):

$$HA = \{R(t_1, ..., t_n) \mid ar(R) = n, t_1, ..., t_n \in GT_L\}$$

Then S' is a set of propositional formulas over the countable set HA.
Moreover, S' has a model if and only if each finite subset of S' has a model (compactness).
A finite subset has a model if and only if propositional resolution does not derive empty clause.
Propositional resolution derives an empty clause iff rules with ground instantiation and resolution derive an empty clause.

Theorem. A set of FOL formulas is unsatisfiable if and only if it is possible to derive empty clause from it using resolution with instantiation.

We can also show: ground instantiation with resolution on ground clauses derives an empty clause if and only iff resolution with unification does (which works better).

A Resolution-Based Prover: **E** by Stephan Schulz

The principles behind Skolemization, transformation to clauses lead to automated theorem provers (ATPs).

Example: E prover. Try E by downloading and building it from:
`https://github.com/eprover/eprover`

To try an example file example.p, use e.g.

```
eprover --auto --proof-object example.p
```

Theorem proving problems, links to competition, other provers:

- `http://www.tptp.org`

Let us Give Our Example to E

Our example in math:

$$\neg\Big(\begin{array}{l}(\forall x.\exists y.\ R(x,y)) \wedge \\ (\forall x.\forall y.\ (R(x,y) \rightarrow \forall z.\ R(x,f(y,z)))) \wedge \\ (\forall x.\ (P(x) \vee P(f(x,a)))) \\ \rightarrow \forall x.\exists y.\ (R(x,y) \wedge P(y))\end{array}\Big)$$

Our example in TPTP ASCII format:

```
fof(ax1,axiom, ![X]: ?[Y]: r(X,Y)).
fof(ax2,axiom, ![X]: ![Y]: (r(X,Y) => ![Z]: r(X,f(Y,Z)))).
fof(ax3,axiom, ![X]: (p(X) | p(f(X,a)))).
fof(c,conjecture, ![X]: ?[Y]: (r(X,Y) & p(Y))).
```

$\wedge$	$\vee$	$\neg$	$\rightarrow$	$\leftrightarrow$	$\forall$	$\exists$
&	|	~	=>	<=>	!	?